5CM522 Operational Research: Weekly Teaching Guide (Lectures)

This guide is built from the restructured repository `ctsallis/operational_research`, under `content/Lectures`. Every statement was checked against the files themselves: slides, sheets, notebooks and staff notes.

For each week it gives:

- **(a) contents:** what each file is for, and whether it is core or optional.
- **(b) teaching order:** the order to use the files in.
- **(c) prerequisites:** what students need beforehand.
- **(d) pitfalls:** known problems.

Where the order comes from the material itself (slide numbers, a session plan, a notebook naming its slide), the guide says *Documented*. Where it is my reading, it says *Judgement*.

Conventions:

- Notebooks run in the JupyterLite site, which uses Pyodide in the browser. The first cell installs Pyomo, SimPy or Mesa where they are needed.
- Pyomo models solve with HiGHS (`appsi_highs`).
- `5 Source Files` is for staff only.
- *Release after the practical* applies to everything in `4 Tutorial Answers`.
- Slide numbers are those of each week's `.pptx` deck. Where a deck has hidden slides, the PDF page numbers used in the week's README and notebooks are lower; each such week says by how much.

Numbers quoted (optimal values, simulation results) match the outputs saved in the notebooks. Every notebook is run in Pyodide, the site's engine, on each push (see *Automatic notebook test* in the repository README).

The audit's fix pack was applied on 12 October 2026, before the current `ctsallis/operational_research` history (which starts on 18 October 2026). It fixed:

- Semester 2 Week 02: the workbook summary formulas.
- Semester 1 Week 05: the slide tables (in the `.pptx`).
- Semester 2 Week 04: the worksheet placement and a file name.
- Smaller fixes in Semester 1 Weeks 03, 07, 10 and 11 and Semester 2 Week 03 (names, notes, comments), plus README notes.

Items marked *(fixed …)* below (12 Oct 2026, or the 15–19 Oct 2026 corrections recorded in each week's README) are kept so you know what changed; the rest still apply.

Semester 1 · Week 01 – Introduction to Operational Research

Handbook topic: *Introduction to Operations Research* (week 1) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 1 - Introduction to Operations Research.pdf`	Lecture slides (30 pages; deck slide 25 is hidden)	Core
2 Lecture Demos	`1 Vectors and matrices in NumPy.ipynb`	Self-paced NumPy primer: vectors, shapes, matrix product, inverse, determinant, solving Ax = b	Core (the tutorial says to do it first)
2 Lecture Demos	`1 Vectors and matrices in NumPy (PDF version).pdf`	Oct 2026 print-out of the current notebook (code only: the notebook stores no outputs)	Optional
3 Tutorial	`Week 1 Tutorial - Introduction to OR.ipynb`	Exercises 1–8: determinants, 2×2 and 3×3 systems by hand and in NumPy, plots, inequalities, interactive 3-D plot	Core

Folder	File	Purpose	Core / optional
3 Tutorial	`Week 1 Tutorial - Introduction to OR (PDF version).pdf`	Oct 2026 print-out of the current tutorial notebook (saved outputs)	Optional
4 Tutorial Answers	`Week 1 Tutorial - Answers.ipynb`	Worked answers to Exercises 1–8: determinants −2, 10, 0; Ex 3 (3, 2); Ex 7 (38/39, 11/39, −4/39)	Core, release after the practical
5 Source Files	`Week 1 - Introduction to Operations Research.pptx`	Editable slides	Staff
5 Source Files	`Week 1 - Tutorial.docx`	Previous MATLAB tutorial, superseded by the notebook	Staff

Tutorial answers were added in Oct 2026 (none in ORIGINAL).

(b) Teaching order

1. Slides 1–3: module handbook, reading and software.
2. Slides 4–14: what OR is, application areas, the two 10-minute pair/group tasks (slides 10 and 13), careers, history video.
3. Slides 15–31 (PDF pages 15–30; deck slide 25 is hidden): maths refresher — linear equations, systems (row/column pictures), matrices, inequalities, feasible regions in 2-D/3-D.
4. Demo: open `1 Vectors and matrices in NumPy.ipynb` after slide 21 (*Matrix*) or at the end of the lecture, and run it top to bottom. *(Judgement: the slides never reference the notebook. The notebook calls itself a short interactive tutorial, so it also works as the first activity of the practical.)*
5. Practical: `Week 1 Tutorial - Introduction to OR.ipynb`, Exercises 1–8 in order. *(Documented: the tutorial's opening Markdown cell says to complete the NumPy notebook first.)*
6. Release `Week 1 Tutorial - Answers.ipynb` after the practical. Exercises 5 and 6 use desmos.com.

(c) Prerequisites

- From earlier study: basic algebra and matrix notation (the slides recap it).
- Open: the JupyterLite site, https://ctsallis.github.io/operational_research/ → `Lectures/Semester 1/Week 01 …`. Run the tutorial's first cell (`%pip install ipywidgets`) before anything else.
- A browser tab with desmos.com for Exercises 5–6.

(d) Pitfalls

- The NumPy notebook's error examples (cells 9, 11, 15) now print the error instead of stopping. If you want students to see a real traceback, delete the `try/except` live.
- The markdown in cell 14 says the browser returns `nan` for a singular matrix instead of raising `LinAlgError`. Check this in JupyterLite before class (the 19 Oct review ran it in Pyodide but does not record which it printed).
- The interactive 3-D plot (Exercise 8) needs `ipywidgets`; if the slider does not redraw in the browser, use Colab.
- The saved tutorial notebook now opens with plots already drawn for the Exercise 4 and 8 example systems (not the exercise answers). Clear outputs if you want a blank sheet.
- *(fixed 19 Oct 2026)* The typo "Write you code here" now reads "Write your code here".

Semester 1 · Week 02 – Linear Programming 1

Handbook topic: *Linear Programming 1* (week 2) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 2 - Linear Programming 1.pdf`	Slides (27): formulation, feasible region, vertices, binding/infeasible/unbounded/multiple optima, canonical form, linprog	Core
1 Lecture	`Simplex Method 2 Variable Graphical Demo.pdf`	12-slide animation of simplex moving A → B → C	Optional ("depending on time", slide 2)

Folder	File	Purpose	Core / optional
2 Lecture Demos	`1 LP with SciPy linprog.ipynb`	Lecture LP (max 5x+7y) in `linprog`: 41.14, 10.29, profit 277.71	Core
2 Lecture Demos	`2 LP with Excel Solver.xlsx`	Same LP set up for Excel Solver	Core (needs desktop Excel)
3 Tutorial	`Week 2 - Linear Programming - Tutorial.pdf`	Q1–Q6	Core
4 Tutorial Answers	`Week 2 - … Tutorial Answers.pdf`	Worked answers	Core, release after the practical
4 Tutorial Answers	`Q3 LP with linprog.ipynb`	Q3: (0, 20), 340	Core
4 Tutorial Answers	`Q4 LP with Excel Solver.xlsx`	Q4: Q1 in the demo workbook's layout, Solver set up; (0, 20), 340	Core (needs desktop Excel)
4 Tutorial Answers	`Q5a/Q5b/Q5c Manufacturing ….ipynb`	Q5: 250; 260; 266.67	Core
4 Tutorial Answers	`Q6 Diet problem - parts a, b and c.ipynb`	Q6: cost 93.25; unbounded; attractiveness 149.66, cost 136.52	Core
5 Source Files	`.pptx`, `.docx`, `Python/*.py`	Editable originals and the scripts behind the notebooks	Staff

(b) Teaching order

1. Slides 1–19: introductory example (two products, one machine), graphical solution, vertex optimality, binding constraints, infeasible / unbounded / infinitely many solutions.
2. Slides 20–24: three-variable picture, canonical (matrix) form, max ↔ min.
3. Slide 25 (*Solution via Python*): run `1 LP with SciPy linprog.ipynb`. *(Documented: the notebook's title cell names this slide.)*
4. Straight after: open `2 LP with Excel Solver.xlsx` and run Solver on the same LP. *(Judgement: the slides do not show Excel, but tutorial Q4 asks for it.)*
5. If time: the simplex graphical demo PDF (slide 2 lists it "depending on time").
6. Slide 27, then the practical: tutorial Q1–Q6. Q6 is flagged as hard and is discussed at the start of Week 3.
7. Release the answer PDF, the Q3–Q6 notebooks and the Q4 workbook after the practical. Hold back the Q6 notebook until after the Week 3 lecture if you want students to attempt it first, because Week 3 demos 2–4 rebuild Q6 in Pyomo.

(c) Prerequisites

- Week 1: matrix form Ax ≤ b, plotting inequalities, NumPy arrays.
- Open the JupyterLite site (Week 02 folder). SciPy is built into the browser Python, so there is no install cell. Desktop Excel with the Solver add-in for the Excel demo.

(d) Pitfalls

- `linprog` minimises: the code uses negative profits and prints `fval = -277.71`. Point out the sign.
- Q5c has tied optima (X and Z both earn £2.667 per machine hour). linprog reports (22.22, 0, 0); the answer sheet now says so and gives the best whole-number plan, £266.
- *(fixed 17 Oct 2026)* The tutorial sheet no longer recommends Spyder: it says Q3, Q5 and Q6 run on the course site and Q4 needs desktop Excel with Solver.
- *(fixed 14 Oct 2026)* The answer sheet's output screenshots are MATLAB output; Q5a now points to the Python notebook. The numbers agree with the Python notebooks.
- The script names (`Question_4a…`, `Question_5`) use last year's numbering; the notebooks use this year's (Q5, Q6).
- *(fixed 14 Oct 2026)* The headers no longer mention Pyomo or GLPK: these notebooks use SciPy only.

Semester 1 · Week 03 – Linear Programming 2

Handbook topic: *Linear Programming 2* (week 3) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 3 - Linear Programming 2.pdf`	Slides (26 pages; deck slide 3 hidden): review, Pyomo, diet problem, intro to (M)ILP	Core
1 Lecture	`Simplex Method 2 Variable Graphical Demo.pdf`	Same simplex animation as Week 2 (slide 2 lists a simplex demonstration)	Optional
2 Lecture Demos	`1 LP with Pyomo.ipynb`	Week 2 lecture LP in Pyomo: 277.71	Core
2 Lecture Demos	`2 Diet problem 5a - minimise cost.ipynb`	Diet LP, minimum cost 93.25	Core
2 Lecture Demos	`3 Diet problem 5b - unbounded.ipynb`	Maximise attractiveness: solver reports *unbounded*	Core
2 Lecture Demos	`4 Diet problem 5c - the fix.ipynb`	Max/min nutrient limits: attractiveness 149.66	Core
2 Lecture Demos	`5 Integer LP.ipynb`	Integer version of the Desmos example: 13 at (3, 2)	Core
2 Lecture Demos	`6 Mixed-integer LP.ipynb`	One integer, one continuous variable: 13.92	Core
3 Tutorial	`Week 3 - Linear Programming 2 - Tutorial.pdf`	Q1 (LP/ILP/MILP), Q2 Icki-Toffee	Core
4 Tutorial Answers	`Week 3 - … Tutorial - Answers.pdf`	Worked answers (with alternative-optimum note on Q2b)	Core, after the practical
4 Tutorial Answers	`Q1 LP, ILP and MILP.ipynb`	270 / 264 / 270 with a feasible-region plot	Core
4 Tutorial Answers	`Q2b …minimise cost.ipynb`, `Q2c … maximise popularity.ipynb`	Cost 268; popularity 365, weight 635 g	Core
5 Source Files	`.pptx`, `.docx`, `Python/`, `Python/Old Q1 …`, `MATLAB/`	Editable sources; old Q1 scripts solve last year's Q1	Staff

(b) Teaching order

(Documented: the slide numbers here are the `.pptx` deck's; the demo notebooks and the Week 03 README give PDF page numbers, which are one lower from slide 4 on because slide 3 is hidden.)

1. Slides 1–6: review of LP, canonical form, `linprog` recap. Optionally show the simplex demo PDF here (slide 2 lists a simplex demonstration).
2. Slide 7: run `1 LP with Pyomo`.
3. Slides 8–12: diet problem formulation (Week 2 Q6, called "5a" on the slides), then run `2 Diet problem 5a`.
4. Slides 13–14: run `3 Diet problem 5b` — unbounded.
5. Slides 15–16: run `4 Diet problem 5c`.
6. Slides 17–23: MILP introduction, manufacturing example, Desmos, rounding.
7. Slide 24: run `5 Integer LP`. Slides 25–26: run `6 Mixed-integer LP`.
8. Slide 27 links to the course site's Week 03 `2 Lecture Demos` folder.
9. Practical: tutorial Q1 (with Q1a, Q1b), then Q2a–d.
10. Release the answer PDF and the three answer notebooks after the practical. Q2d ("is this a good selection?") is discussed at the start of Week 4.

(c) Prerequisites

- Week 2: LP formulation, `linprog`, and especially Q6 (diet problem), which this lecture solves.
- Open: JupyterLite, Week 03 folder. Run each notebook's first cell (installs Pyomo and HiGHS; allow a minute the first time).

(d) Pitfalls

- The slides call the diet problem "Tutorial Question 5" and the manufacturing problem "Q4"; on this year's Week 2 sheet they are Q6 and Q5.
- Q2b has several cheapest selections. HiGHS gives (x_1, x_9, x_{11}) = (9, 8, 4); the answer sheet shows (9, 10, 3). Both cost 268, and the sheet now says any 268 selection is correct.
- Demo 3 is meant to be unbounded. It prints the termination condition instead of a solution; don't try `model.display()` after it.
- Do not mix SciPy `linprog` and Pyomo/HiGHS in the same browser notebook (the root README says this crashes the kernel).
- *(fixed 12 Oct 2026)* Demos 1 and 6 had the comment `# Solve with GLPK` above `SolverFactory('appsi_highs')`; it now says HiGHS.

Semester 1 · Week 04 – Mathematical Programming: Modelling

Handbook topic: *Mathematical Programming – Modelling and Some Issues* (week 4) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 4 - Mathematical Programming - Modelling.pdf`	Slides (26 pages; exported from the .pptx for NEW): ≤ vs <, summation/matrix form, Pyomo expressions, chocolate-box (Icki-Toffee) results, ratio/weighted/normalised objectives, modelling tips	Core
1 Lecture	`Slack Variables.pdf`	10-slide graphical demo of slack variables and binding/non-binding constraints; last slide: slack vs surplus	Core (short)
2 Lecture Demos	`1 Pyomo expressions and constraints.ipynb`	Builds and prints the expression and constraint shown on slides 6–7 (PDF pages 5–6)	Core
3 Tutorial	`Week 4 - … - Tutorial.pdf`	Q1 Icki-Toffee (weight cap, combined objective), Q2 two factories	Core
4 Tutorial Answers	`Week 4 - … - Tutorial - Answers.pdf`	Worked answers; Q1b points to the notebooks and tabulates their selections	Core, after the practical
4 Tutorial Answers	`Q1b … weighted / normalised weighted / popularity per cost ratio.ipynb`	Three ways to combine cost and popularity (results match slide 24)	Core
4 Tutorial Answers	`Q2a Factory A.ipynb`, `Q2a Factory B.ipynb`, `Q2b Factories combined.ipynb`	220, 165, combined 400	Core
5 Source Files	`.pptx`, `.docx`, `Python/`, `MATLAB/`	Sources	Staff
6 Further Reading	Woolsey "St. Jude" papers (×2)	Recommended on the last slide	Optional

(b) Teaching order

(Slide numbers are the `.pptx` deck's; slide 3 is hidden, so the README and notebooks give PDF pages one lower from slide 4.)

1. Slides 1–5: closed feasible regions (≤ not <), summation and matrix form. *(Documented: the README puts `Slack Variables.pdf` after slide 27, before the practical, as Q2a ii needs it; it is a separate deck, not referenced by the main slides.)*
2. Slides 6–7: run `1 Pyomo expressions and constraints.ipynb`. *(Documented in the notebook's first cell, as PDF pages 5–6.)*
3. Slides 8–10: diet problem recap (Week 3 demos 2–4 can be reused).

4. Slides 11–17: the chocolate-box results from Week 3 Q2, interpretation, opinion, and the weight cap (tutorial Q1a).
5. Slides 18–24: ratio, weighted and normalised objectives. Show the three Q1b notebooks *after* the practical, as they are the Q1b answers.
6. Slides 25–27: solve the LP first, a modelling recipe, tips; point students to the Woolsey papers.
7. Practical: Q1a, Q1b, then Q2a, Q2b.
8. Release the answers PDF and the six answer notebooks after the practical.

(c) Prerequisites

- Week 3: Pyomo basics and the Icki-Toffee model (Q2); the cost-minimising model is the starting point for Q1.
- Open: JupyterLite Week 04 folder; run each notebook's first cell.

(d) Pitfalls

- Q1b weighted objective: the optimum (305) is not unique. HiGHS gives Coconut 30 / Rose 5 / Grape 5 (as on slide 24); GLPK gives 29 / 2 / 4 / 5. The notebook explains this.
- The weighted notebook uses `alpha = 0`, which makes it a pure popularity model. Try other α values live (weighted: $\alpha \gtrsim 0.42$ reproduces min-cost, $\alpha \lesssim 0.09$ max-popularity; slide 24's 0.59 and < 1/6 are for the normalised objective).
- *(fixed 19 Oct 2026)* The hidden filler terms in the answer sheet's Q2b formulation were removed; each constraint now shows only its own factory's variables.
- *(fixed 17 Oct 2026)* The answer sheet's Colab link now points to the answer notebooks on the course site.
- *(fixed 17 Oct 2026)* The lecture PDF was re-exported from the corrected `.pptx` in PowerPoint; its table headers now show correctly.

Semester 1 · Week 05 – Mixed Integer Programming

Handbook topic: *Mixed Integer Programming* (week 5) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 5 - Mixed Integer Programming.pdf`	Slides (29 pages; deck slides 3, 22, 23 hidden): MILP review, rounding, binary variables, knapsack, set covering, purchasing	Core
2 Lecture Demos	`1 Integer LP`	P = 2x + 3.5y: 13 at (3, 2)	Core
2 Lecture Demos	`2 Knapsack problem`	7 items, 10 kg: value 170	Core
2 Lecture Demos	`3 Fire stations - set covering`	6 cities: build at 2 and 4	Core
2 Lecture Demos	`4–7 Purchasing …`	Basic 84.44; binary min-order 81.6; at most two items 80; item 1 or 4 76.67	Core
2 Lecture Demos	`8 Minimum vertex cover`	Small vertex-cover example (6 vertices, 5 edges; minimum cover 3)	Optional — not in the Week 5 slides; fits Week 6
3 Tutorial	`Week 5 - Mixed Integer Programming Tutorial.pdf`	Q1 desert-island knapsack, Q2 fire stations, Q3 Icki-Toffee with binaries	Core
4 Tutorial Answers	answers PDF, `Q1a/Q1b/Q1c`, `Q2`, `Q3` notebooks	270 / 350 / 250; 3 stations; cost 310	Core, after the practical
5 Source Files	`.pptx`, `.docx`, `Python/`, `MATLAB/`	Sources	Staff

(b) Teaching order

(Documented: each notebook's first cell and the README give the positions as PDF pages. The slide numbers below are the `.pptx` deck's, which hides slides 3, 22 and 23: deck slides 4–21 are PDF pages 3–20, and 24–32 are pages 21–29.)

1. Slides 1–8: MILP review and rounded solutions.
2. Slide 9 (*Integer Solution via Pyomo*): run `1 Integer LP`.
3. Slides 10–13: binary variables, knapsack formulation. Slide 14: run `2 Knapsack problem`.
4. Slides 15–20: fire-station set covering (slide 16: "build for city 2 and 4"). Slide 21: run `3 Fire stations`.
5. Slides 24–25: purchasing basic model, run 4. Slides 26–28: binary min-order, run 5. Slides 29–30: at most two items, run 6. Slides 31–32: item 1 or item 4, run 7.
6. Optional: `8 Minimum vertex cover` as a preview of Week 6. *(Judgement.)*
7. Practical: Q1a–c, Q2, Q3. Release the answers after the practical.

(c) Prerequisites

- Weeks 3–4: Pyomo models; the Icki-Toffee model with the weight cap (Week 4 Q1a) is the base for Q3.
- Open: JupyterLite Week 05 folder; run the first cell of each notebook.

(d) Pitfalls

- The purchasing tables on slides 26–31 showed "Expected Profit" equal to cost (10, 12, 9, 5, 13); the code and slide 25 use 5, 3, 4, 3, 3. *(fixed 12 Oct 2026 in the `.pptx`)* **The lecture PDF in `1 Lecture` was re-exported in PowerPoint (latest 19 Oct 2026) and shows the corrected values.**
- Several problems have more than one optimum (knapsack demo, Q2 fire stations, Q3 Icki-Toffee, vertex cover). HiGHS may show a different selection from the MATLAB screenshots; the optimal value is the same. The answer sheet now says so for Q2 and Q3.
- The answer screenshots are MATLAB output (`sol = …`, `fval = -270`, negative because MATLAB minimises).

Semester 1 · Week 06 – Integer Linear Programming: Further Problems

Handbook topic: *Integer Linear Programming – Further Problems* (week 6) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 6 - … Further Problems.pdf`	Slides (25 pages; deck slides 2, 10, 11 hidden): set cover, costed fire stations, vertex cover, matching, balanced assignment	Core
1 Lecture	`… (annotated).pdf`	The 2023 lecture with handwritten notes (28 pages, includes hidden slides; older notation)	Optional
2 Lecture Demos	`1-3 Costed fire stations …`	Same model three ways (plain, incidence matrix, sets/dictionaries): cost 24	Core
2 Lecture Demos	`4 Security cameras - vertex cover`	8-node corridor graph: 4 cameras	Core
2 Lecture Demos	`5-7 Balanced assignment …`	4×4 assignment three ways: cost 13	Core
4 Tutorial Answers	`Q5 Maximum set cover, edge cover and independent set.ipynb`	Task 5 on slide 28: 4 cities with two stations, 4 corridors, 4 corners	Core, after the practical
5 Source Files	`.pptx` (×2), `Python/`, `Staff Notes/Notes for 2526.txt`	Sources; staff note "Include Pyomo Modelling"	Staff
6 Further Reading	`How-Vodafone-Uses-Optimization.pdf`	Industry case study	Optional

There is no tutorial sheet; slide 28 (PDF page 25) sets the tutorial tasks, and `4 Tutorial Answers` holds the task 5 answer.

(b) Teaching order

(Documented: the README and notebooks give the positions as PDF pages. The numbers below are `.pptx` slides: PDF page + 1 for slides 3–9, + 3 from slide 12.)

1. Slides 1–6: general set covering.
2. Slides 7–9: costed fire stations; run `1 Costed fire stations` (cost 24 = slide 9), then 2 and 3 to show the incidence-matrix and sets/dictionaries styles.
3. Slides 12–16: vertex cover; slides 17–18: run `4 Security cameras` and compare its cover with the two shown on the slides.
4. Slides 19–26: matching and balanced assignment; at slide 26 run 5, then 6 and 7 for named indices.
5. Slide 27: dissertation assignment exercise (discussion).
6. Slide 28: tutorial — study the code, recode the costed fire station with an incidence matrix (demo 2), code a Gurobi example and the task 5 problems in Pyomo; then release the `Q5 …` notebook.
7. Further reading: the Vodafone case study.

(c) Prerequisites

- Week 5: binary variables and set covering (fire stations). Optionally Week 5 demo 8 (vertex cover) as a warm-up.
- Open: JupyterLite Week 06 folder.

(d) Pitfalls

- Vertex cover has exactly two optimal covers, both on the slides: {2, 4, 6, 8} ("MATLAB Solution", also HiGHS's pick) and {1, 3, 5, 7} (the GLPK script's pick). Both use 4 cameras (the notebook says so).
- Numbering varies: demo 3 numbers cities 1–5 (the slides use 0–4), demos 5–6 index workers from 0 (workers 1–4 → indices 0–3), and demo 7 from 1; each notebook says so. Point this out when reading the output.
- The annotated PDF (2023) uses older notation; only the annotated `.pptx` in `5 Source Files` (2020) refers to MATLAB. Never export it over the PDF.

Semester 1 · Week 07 – Flow Shop and Job Shop Scheduling

Handbook topic: *Scheduling* (week 7) — matches. The folder holds what used to be two sessions (ORIGINAL sub-folders "Week 1 - Flow Shop" and "Week 2 - Job Shop", from when this was Semester 2 weeks 1–2), so plan for two lectures' worth of material or choose.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 7 - Flow Shop Scheduling.pdf`	25 slides: SPT, flow shop, makespan/idle time, Johnson's rule, >2 machines	Core
1 Lecture	`Week 7 - Job Shop Scheduling Notes.pdf` and `Notes/Job Shop Scheduling Notes (Short) (.md/.ipynb)`	Job-shop notes: disjunctive MILP in Pyomo, example makespan 11	Core
1 Lecture	`Notes/Job Shop Scheduling Notes (.md/.ipynb)`	Full notes: adds constraint programming and cryptarithmetic	Optional
2 Lecture Demos	`1 Johnsons rule.ipynb`	Johnson's rule (recursive and iterative) on the slide examples	Core
2 Lecture Demos	`2 Johnsons rule in Excel.xlsx`	Worked examples 1–2, >2 stages, Gantt charts, tutorial sheet	Core
3 Tutorial	`Week 7 - Flow Shop Scheduling - Tutorial.pdf`	Flow-shop Q1–Q4	Core
3 Tutorial	`Week 7 - Job Shop Scheduling Tutorial.pdf`	Job-shop Q1–Q3 (+ Q4–Q5 reading on OR-Tools)	Core
4 Tutorial Answers	`Week 7 - Flow Shop Scheduling - Tutorial - Answers.pdf`	Makespan 30 (Q1), 74 (Q2), Q3 heuristic 66, optimum 64	Core, after the practical
4 Tutorial Answers	`Flow Shop Q4 MILP answers.ipynb`	Flow-shop Q4: permutation MILP, makespans 30, 74, 64	Core, after the practical

Folder	File	Purpose	Core / optional
4 Tutorial Answers	`Job Shop Q1-Q3 answers.ipynb`	Makespans 12, 16, 18	Core, after the practical
5 Source Files	sources, HTML exports, `Previous Versions/`	Last year's slides and notes	Staff
6 Further Reading	Bockmayr & Reinert; Rodríguez-Carbonell	Integer vs constraint programming	Optional

(b) Teaching order

Session A – flow shop *(slide order and demo placement documented in the README)*

1. Slides 1–12: single-stage scheduling (SPT), flow shop, makespan and idle time, why optimality is hard.
2. Slides 13–18: Johnson's rule and Example 1 (40 → 30). Run `1 Johnsons rule.ipynb`, then open the Excel `Example 1/Gantt 1` sheets.
3. Slides 19–20: Example 2 (19 → 18), Excel `Example 2`.
4. Slides 21–24: more than two machines (Excel `More than 2 stages`).
5. Practical: flow-shop tutorial Q1–Q4; Q3 suggests Excel (`Tutorial` sheet). Release the flow-shop answers and `Flow Shop Q4 MILP answers.ipynb`.

Session B – job shop *(judgement: the order follows the notes)*

6. Work through `Job Shop Scheduling Notes (Short)` (or its PDF), running the notebook's cells as you go (example makespan 11); the full notes are optional.
7. Practical: job-shop Q1–Q3 (adapt the notes' model). Q4–Q5 are reading exercises on the Google OR-Tools site and need Colab.
8. Release `Job Shop Q1-Q3 answers.ipynb`. Further reading: the two constraint-programming PDFs.

(c) Prerequisites

- Weeks 5–6: binary variables and big-M style constraints (the job-shop model uses disjunctive constraints).
- Open: JupyterLite Week 07 folder; desktop Excel for the workbook; Google Colab for job-shop Q4–Q5.

(d) Pitfalls

- The job-shop tutorial still opens with Google Colab and OR-Tools, but now says Q1–Q3 can be done in the browser with the notes' Pyomo code; the answers are Pyomo (OR-Tools cannot run in the browser).
- Optimal makespans are unique but schedules are not; students' start times may differ from the notebook's.
- *(fixed 14 Oct 2026)* The old OR-Tools answer script, which did not run and had the wrong Q1 data, has been removed.
- *(fixed 12 Oct 2026)* The annotated flow shop slides in `Previous Versions` are dated 2021 and are now named `(2020-21 slides, annotated)`.
- *(fixed 14 Oct 2026)* The Johnson's-rule notebook converts job numbers with `int()`, so the browser prints `0` rather than `np.int64(0)`; the orders are unchanged.

Semester 1 · Week 08 – Networks: Shortest Paths and MST

Handbook topic: *Network Models: Transportation and Transhipment Problem* (week 8) — **mismatch**. This folder teaches shortest paths and minimum spanning trees. Transhipment/min-cost flow examples exist in Semester 2 Week 01 (`6 Production-distribution (minimum-cost flow)`) and Week 10 (`F.B. Knight - transhipment`), as the README notes.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 8 - Networks - Shortest Paths and MST.pdf`	70 slides: networks, adjacency matrix, Dijkstra worked example, correctness, Prim, Kruskal	Core

Folder	File	Purpose	Core / optional
1 Lecture	… (annotated).pdf	Annotated version from a previous delivery	Optional
1 Lecture	Dijkstra Algorithm.pdf	Third-party deck (B. C. Dean, Clemson) on shortest paths, Bellman-Ford, Dijkstra	Optional (extra depth)
1 Lecture	Negative weight example (Dijkstra fails).pdf	6-slide counter-example: Dijkstra with a negative edge	Core (short)
3 Tutorial	Week 8 - Tutorial - Networks - Shortest Paths and MST.pdf	Q1 MST (Prim and Kruskal); Q2a/b Dijkstra	Core
4 Tutorial Answers	… Q1 Answers.pdf, … Q2 Answers.pdf; Q1-Q2 answers in Python.ipynb	Step-by-step slides: MST weight 39; 1→4 = 5; 1→2 = 55. The notebook runs Dijkstra, Prim and Kruskal in plain Python	Core, after the practical (notebook optional)
5 Source Files	.pptx ×5, .docx, Staff Notes/	Sources; empty note "Do flow shop scheduling problem live"	Staff
6 Further Reading	MIT 6.006 Lecture 16 - Dijkstra.pdf	MIT lecture notes	Optional

No lecture notebooks: this week is worked by hand; an optional answers notebook checks the steps in plain Python.

(b) Teaching order

(Slide order documented; the placement of the two extra decks is judgement.)

1. Slides 1–6: networks, adjacency matrix, shortest-path definition, "Blue Peter" example.
2. Slides 7–31: Dijkstra set-up, cost labels, the step-by-step example (slides 11–28), correctness and complexity.
3. Straight after: Negative weight example (Dijkstra fails).pdf, why Dijkstra needs non-negative weights.
4. Slides 32–35: break activity.
5. Slides 36–55: minimum spanning tree, fibre-optic application, Prim's worked example.
6. Slides 56–69: Kruskal's worked example; slide 70 references.
7. Optional: Dijkstra Algorithm.pdf (Bellman-Ford, centrality) and the MIT notes for further depth.
8. Practical: Q1 (Prim and Kruskal), Q2a, Q2b. Release both answer PDFs and Q1-Q2 answers in Python.ipynb after the practical.

(c) Prerequisites

- No coding prerequisite. Graph vocabulary is introduced here; Week 6 (vertex cover, matching) used graphs informally.
- Paper and pen; the answer PDFs are animations of the tables, so show them in slide-show mode if possible.

(d) Pitfalls

- *(fixed 12 Oct 2026)* In the negative-weight example, the last slide's table said node 4 = −3; it now says −2, as the graph and the reasoning give.
- Q2a has two shortest paths of length 5 (1→3→4 and 1→2→3→4); both are correct.
- *(fixed 17 Oct 2026)* The Q2 answers PDF was re-exported from the .pptx; its 14 pages match the 15-slide deck, whose slide 10 (an unlabelled copy of the Q2b start) is hidden.
- The empty staff note in this folder refers to flow-shop scheduling (Week 7).

Semester 1 · Week 09 – Travelling Salesman Problem

Handbook topic: *Travelling Salesman Problem* (week 9) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	Travelling Salesman Problem.pdf	23 slides: complexity, applications, MILP, subtours, MTZ, heuristics	Core

Folder	File	Purpose	Core / optional
2 Lecture Demos	`1 TSP with Pyomo.ipynb`	MTZ model on 4 and 20 random cities (seeded), plots; a genetic algorithm on 35 cities	Core (also the tutorial)
4 Tutorial Answers	`Tutorial answers (slide 22).ipynb`	Answers to slide 22: solve times for 5–35 cities, five genetic-algorithm settings vs the optimum 518.52, research questions	Core, after the practical
5 Source Files	`Travelling Salesman Problem.pptx`	Editable slides	Staff

No tutorial sheet: slide 22 makes the notebook the tutorial; the answers are in `4 Tutorial Answers`.

(b) Teaching order

(Slide order documented; slides 20 and 22 point to `1 TSP with Pyomo.ipynb` on the course site, which is this file.)

1. Slides 1–11: the problem, complexity, the $1M prize, applications, Bill Cook.
2. Slides 12–19: MILP formulation, subtours, why the constraints work, MTZ.
3. Slide 20 (*Pyomo Code*): run the first three code cells of `1 TSP with Pyomo.ipynb` (4-city tour = 80; 20-city tour = 407.97, with plots).
4. Slide 21: heuristics — run the genetic-algorithm cell and compare.
5. Slide 22: practical — students vary `num_cities` and explore; release `Tutorial answers (slide 22).ipynb` afterwards. Slide 23: video.

(c) Prerequisites

- Weeks 5–6: binary variables; Week 8: networks.
- Open: JupyterLite Week 09 folder; run the first cell. The 20-city model has 419 variables; allow a little time in the browser.

(d) Pitfalls

- *(fixed 14 Oct 2026)* The genetic algorithm is now seeded (`random.seed(1)`), so its "Best fitness" is the same on every run (599.24). Change or delete the seed to show the variation.
- *(fixed 17 Oct 2026)* Slides 20 and 22 said the notebook was on Blackboard/Colab; they now point to `1 TSP with Pyomo.ipynb` on the course site.
- Larger `num_cities` values make the MILP slow in the browser.

Semester 1 · Week 10 – Vehicle Routing Problem

Handbook topic: *Vehicle Routing Problem* (week 10) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Vehicle Routing Problem.pdf`	35 pages (deck slide 2 hidden): TSP/MTZ recap, multiple tours, VRP min total distance (flow formulation), min longest route	Core
1 Lecture	`Lecture Notes - Routing Problems.pdf` / `Notes/Routing Problems - Lecture Notes.md`	Full written notes with Pyomo code and figures	Core (reading)
2 Lecture Demos	`1 TSP and VRP with Pyomo.ipynb`	New companion: one 8-node example, all three models, route plots, "Try it" exercises	Core
5 Source Files	`.pptx`, HTML exports, `Previous Versions/`, `Staff Notes/` (slides 27 and 33)	Sources	Staff
5 Source Files	`Previous Versions/Routing - Short Notes for the Lecture (earlier draft).md`	Earlier, shorter draft; its minimax model is wrong	Staff
6 Further Reading	Miller–Tucker–Zemlin (1960); Christofides (1976)	Original papers	Optional

No tutorial sheet; the notebook's "Try it" cell (hidden answers, also in the README) is the practical, and the last slide links to it on the course site.

(b) Teaching order

(Slide order documented; demo placement matches the notebook's three sections to the slides.)

1. Slides 1–13: TSP reminder, MTZ constraints, set notation. Run notebook section 1 (TSP with/without MTZ: subtours at 233.4, tour 281.4).
2. Slides 14–17: multiple tours from the depot.
3. Slides 18–28: VRP definition, objective, demand/capacity, flow formulation, why no subtour constraints. Run section 2 (minimum total distance 327.6, two vehicles used).
4. Slides 29–33: minimise the longest route. Run section 3 (longest route 137.4, total 362.5).
5. Slides 34–35: subgraph variant, conclusion.
6. Practical: notebook "Try it" questions (capacity 12, two vehicles, extra customer, cost of balancing). Slide 36 links to this notebook on the course site.
7. Reading: lecture notes, then the MTZ and Christofides papers.

(c) Prerequisites

- Week 9: the TSP and MTZ formulation (re-used directly).
- Weeks 5–6: binary variables; Week 8: networks.
- Open: JupyterLite Week 10 folder; the min-longest-route model has 337 variables, so allow time in the browser.

(d) Pitfalls

- *(fixed 17 Oct 2026)* Slide 36 linked to a Google Colab notebook that is not in the repository; it now links to the companion notebook on the course site.
- The minimum-total-distance model may use fewer vehicles than available (two of three here). This is intended; slide 23 discusses it.
- *(fixed 12 Oct 2026 in the `.md` notes)* The MTZ model had placeholder equation labels ("dsad"). The HTML exports in `5 Source Files` were corrected on 14 Oct 2026.
- *(fixed 14 Oct 2026)* The earlier HTML export with broken image links was removed.

Semester 1 · Week 11 – Floor Planning

Handbook topic: *Inventory Modelling* (week 11) — **mismatch**. This folder holds floor planning / bin packing (as the README notes). No Inventory Modelling material exists in either repository.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Week 11 - Floor Planning.pdf`	35 pages (deck slide 2 hidden): LP/MILP recap, 2-D floor planning, non-overlap constraints, rotation, connected packing, "Mums know best"	Core
1 Lecture	`Notes/Floor Planning - Pyomo Model.md`	The Pyomo model of demo 1 in one piece, to read (uses GLPK)	Core (reading)
2 Lecture Demos	`1 Floor planning - minimise area.ipynb`	Minimum square for 6 items with rotation: side 12	Core
2 Lecture Demos	*(added 12 Oct 2026)* `2 Floor planning - maximise number of items.ipynb`	Port of the MATLAB max-items variant: 6 of 7 items on a 4×5 floor	Optional extra
2 Lecture Demos	*(added 12 Oct 2026)* `3 Floor planning - maximise placed area.ipynb`	Port of the MATLAB max-area variant: 100 on a 10×10 floor	Optional extra

Folder	File	Purpose	Core / optional
5 Source Files	`.pptx`, `MATLAB/` (3 files), `Python/model.py`, HTML export	Sources	Staff

(b) Teaching order

(Slide order documented; demo positions by matching the model to the slides. Slide numbers are the `.pptx` deck's; the README and notebooks use PDF pages, one lower from slide 3.)

1. Slides 1–5: LP/MILP recap.
2. Slides 6–19: the 2-D floor-planning model: positions, four non-overlap cases, binary variables, minimax objective.
3. Slide 20 (*Solution*, A* = 12 for 4 and 6 items): run `1 Floor planning - minimise area` with the `rotation_off` constraint uncommented (no rotation).
4. Slides 21–24: adding rotation; re-run the demo with rotation allowed (6 items: still 12; the slide's 11 is for the 4-item set in the commented data line).
5. Optional: notebooks 2 and 3 (fixed floor; maximise items or area) as a modelling variation; notebook 2 needs 2–4 minutes in the browser. *(Judgement — not in the slides.)*
6. Slides 25–32: connected packing (distance objective); slides 33–36: complexity, "Mums know best", conclusion.
7. Reading: the Pyomo-model notes. There is no tutorial sheet this week.

(c) Prerequisites

- Weeks 5–7: binary variables, big-M/disjunctive constraints (job-shop scheduling uses the same either/or idea).
- Open: JupyterLite Week 11 folder; models have 49–70 variables; demos 1 and 3 solve in seconds, demo 2 takes 2–4 minutes.

(d) Pitfalls

- Big-M = 9999 (as in the MATLAB files and the min-area demo) only causes rounding: a desktop prints `v= 11.999999`. The ports use a tight M as good practice; the earlier false optimum of 5 items with M = 9999 was not reproduced (checked Oct 2026).
- The demo's plot treats `model.r[i].value` as true/false. A solver value such as 1e-10 would draw an item as rotated; test `> 0.5`.
- Many layouts share the optimal area, so the drawing can differ from the slides.
- *(fixed 14 Oct 2026)* The HTML export in `5 Source Files` no longer links a stylesheet on the author's PC and now uses `plt.get_cmap`, like the `.md` notes.

Semester 2 · Week 01 – Introduction to Simulation

Handbook topic: *Introduction to Simulation* (S2 week 1) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Introduction to Simulation.pdf`	39 pages (deck slides 3, 22 hidden): why simulate, coins and dice, queueing definitions, operating a simulation by hand	Core
1 Lecture	`Example Simulation.pdf`	One-page hand-simulation handout (the slides 38–40 example)	Core
1 Lecture	`Production Distribution Model.pdf`	11-slide minimum-cost-flow (production/distribution) example	Optional here — it is an optimisation topic (fits S1 Week 8)
2 Lecture Demos	`1 Coin toss simulation.ipynb`, `2 Dice simulation.ipynb`	Law of large numbers; Python versions of the MATLAB demos	Core

Folder	File	Purpose	Core / optional
2 Lecture Demos	`3 Coin and dice simulations in Excel.xlsx`	The Excel runs shown on slides 13–15	Core
2 Lecture Demos	`4 Heated clothes rack in Excel.xlsx`	Slide 11 example	Core
2 Lecture Demos	`5 Election simulation.ipynb`	Monte Carlo election (2020 polls)	Optional (not in the slides)
2 Lecture Demos	`6 Production-distribution (minimum-cost flow).ipynb`	Solves the production-distribution deck's model: £533,930	Optional (goes with that deck)
3 Tutorial	`Tutorial - Introduction to Simulation.pdf`, `Tutorial Example.xlsx`, `Tutorial Excel Sheets.xlsx`	Q1 by hand, Q2–Q3 in Excel, Q4 Python dice	Core
4 Tutorial Answers	answers PDF, `Tutorial Answers.xlsx`, `Tutorial example answer.png`, `Tutorial Q2-Q4 in Python.ipynb`	Worked answers (Excel; handwritten Q1 table; Q2–Q4 in Python)	Core, after the practical
5 Source Files	`.pptx`, `.docx`, `MATLAB/` (4 files)	Sources; MATLAB originals of the four notebooks	Staff

(b) Teaching order

(Slide order documented; demo placement by matching each demo to the slide that shows the same experiment. Slide numbers are `.pptx` deck numbers; the README uses PDF pages, one lower from slide 4 and two lower from slide 23.)

1. Slides 1–10: what simulation is, approaches, stages, why simulate.
2. Slide 11: open `4 Heated clothes rack in Excel.xlsx`.
3. Slides 12–15: coins and dice. Open `3 Coin and dice simulations in Excel.xlsx` (the slides' screenshots), then run `1 Coin toss` and `2 Dice` to show the same thing in Python. Press F9 in Excel to re-roll.
4. Slides 16–27: queueing vocabulary, examples (doctor's waiting room, traffic lights).
5. Slides 28–40: operating a simulation by hand; hand out `Example Simulation.pdf` for slides 38–40.
6. Slide 41: tips (warm-up, run length).
7. Optional: `5 Election simulation` as a Monte Carlo extension; the production-distribution deck and notebook if transhipment was not covered in S1 Week 8. *(Judgement.)*
8. Practical: tutorial Q1 (by hand), Q2–Q3 in `Tutorial Excel Sheets.xlsx` (worked example in `Tutorial Example.xlsx`), Q4 Python dice. Release the answers after the practical, including `Tutorial Q2-Q4 in Python`. The `2 Dice simulation` notebook also answers Q4a.

(c) Prerequisites

- Basic probability (relative frequency, cumulative probability).
- Excel (random numbers, LOOKUP with cumulative probabilities); JupyterLite for the notebooks (NumPy/Matplotlib/pandas; only demo 6 has an install cell, for Pyomo + HiGHS).
- For the production-distribution example: S1 Weeks 2–3 (LP) and Pyomo.

(d) Pitfalls

- Every Excel workbook except the heated clothes rack uses `RAND()`: values change on every recalculation, so students' numbers will not match the answer workbook exactly. Compare shapes and long-run percentages, not individual cells.
- The coin notebook plots percentages (0–100); the MATLAB original and some slides use proportions (0–1).
- Notebook seeds are fixed (`default_rng(1)`), so every student gets the same run. Remove the seed to show variability.
- The election demo uses 2020 US polling data. It is not in the slides; introduce it as an extra.
- The PDF (slide 39, p. 37) shows the inter-arrival band for 12 as 0.2–0.4; it should be 0.2–0.6 (fixed in the `.pptx`). In `Tutorial Excel Sheets.xlsx` (*Uniform Arrivals and Service*) `INT(RAND()*(max–min)+min)` never reaches max; use `RANDBETWEEN`.

Semester 2 · Week 02 – Queueing Theory

Handbook topic: *Queuing Theory* (S2 week 2) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Introduction to Queueing Theory.pdf`	25 pages (deck slide 3 hidden): distributions, Little's theorem, Kendall notation, M/M/1 formulas and cost example, disciplines, servers in series	Core
1 Lecture	`M-M-1 queue derivation.pdf`	8-page notes deriving the M/M/1 results	Optional (reading)
2 Lecture Demos	`1 M-M-1 lecture example.xlsx`, `2 Queueing examples in Python.ipynb`	Excel M/M/1 simulation; Python versions of the slide examples (M/M/1, costs, servers in series, theory vs simulation)	Core
3 Tutorial	`Tutorial - Queueing Theory.pdf`, `Tutorial Template.xlsx`	Q1 M/M/1 bank (λ = 1/3, μ = 2/3 per min), Q2 normal service, Q3 servers in series, Q4 Q1–Q3 in Python	Core
4 Tutorial Answers	`Tutorial Answers.xlsx`, `Tutorial answers in Python.ipynb`	Worked simulations and theory (Wq = 1.5, W = 3 for Q1); Q4 in Python	Core, after the practical
5 Source Files	`.pptx`, `.docx`	Sources	Staff

Two notebooks (NumPy/Matplotlib, no install cell): the slide examples and the Q4 answers. SimPy arrives in Week 03.

(b) Teaching order

(Slide order documented; demo placement judgement — slides 16–17 show results of this kind of spreadsheet; slides 19–26 have no workbook, and demo 2 covers them. Slide numbers are `.pptx` deck numbers; the README and demo 2 use PDF pages, one lower from slide 4.)

1. Slides 1–11: why queues, arrival/service distributions (Poisson, exponential, uniform, normal).
2. Slides 12–15: Little's theorem, Kendall notation, M/M/1 formulas.
3. Slides 16–22: M/M/1 example and costs. Open `1 M-M-1 lecture example.xlsx` to show simulated vs theoretical values, then run `2 Queueing examples in Python` sections 1–3.
4. Slides 23–26: disciplines and servers in series (`2 Queueing examples in Python`, section 4).
5. Reading: `M-M-1 queue derivation.pdf`.
6. Practical: Q1–Q3 using `Tutorial Template.xlsx`, Q4 in Python. Release `Tutorial Answers.xlsx` and `Tutorial answers in Python` afterwards.

(c) Prerequisites

- S2 Week 01: hand/Excel simulation, `RAND()` and lookup of random variates, queueing vocabulary.
- Desktop Excel (or LibreOffice). Basic exponential/normal distributions.

(d) Pitfalls

- *(fixed 12 Oct 2026)* Both the template and the answers had wrong summary formulas (carried over from ORIGINAL):
  - The cells labelled "Average Arrival Rate (computed from IAT)" and "Average Service Rate (computed from Service Time)" (Q1 O3/O4, Q3 V3/V4) averaged the *other* column and gave mean times, not rates, over rows 2–356 only. They are now `1/AVERAGE(...)` of the right column over all 1000 rows.
  - Q3's "Average Time in Queue/System" (V5/V6) averaged the running-average columns; they now average the per-customer times (2.49 / 12.44 on the saved run, was 2.22 / 12.19).
  - In the template, V4–V6 show `#DIV/0!` until students fill in their columns. This is expected.
- Results change on every recalculation (`RAND()`); compare with the theoretical values, not with the answer workbook's exact numbers.

- Keep units consistent (the sheet works in minutes: λ = 20/60, μ = 40/60; the template opens with placeholders λ = 1, μ = 10, so students must enter them).
- The large workbooks (0.9–1.3 MB, ~40,000 formulas) recalculate slowly.
- The cost example (slides 20–22) is one run: μ = 4 and μ = 2 cost almost the same on average (£91.2 vs £91.8), so 'case 2 is cheapest' is not general; demo 2 shows both. The derivation notes have typos on pp. 2, 5 and 6 (README errata).

Semester 2 · Week 03 – Discrete Event Simulation with SimPy

Handbook topic: *Discrete-Event Simulation* (S2 week 3) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Simpy Basics.pdf`	One-slide diagram of a customer moving through a SimPy resource (slide 1 blank; the PDF shows the animation's final state)	Core (short)
1 Lecture	`Notes/M-M-1 queue - three snapshots.md`	Short illustration of queue states	Optional
2 Lecture Demos	`1–3` segment notebooks	Runnable versions of the three segments	Core
2 Lecture Demos	`4 Machine breakdowns - full demo.ipynb`	Full breakdown/maintenance demo (two MTBF scenarios)	Optional (extends segment 3)
3 Tutorial	`Practical - M-M-1 Queue.ipynb` / `.pdf`	Guided M/M/1 practical	Core
3 Tutorial	`Tutorial Questions.pdf`	Q1–Q5 (bank, supermarket, call centre, CNC, hospital ED)	Core
4 Tutorial Answers	`Q1–Q5` notebooks	Worked SimPy answers	Core, after the practical
5 Source Files	`.pptx`, `Notes (Markdown)/`, `Python/`, `Staff Notes/`, `Previous Versions/` (one earlier HTML draft)	Sources; `Session Outline.md` has minute-by-minute timings	Staff
5 Source Files	`Notes (Markdown)/1 …`, `2 …`, `3 …` (.md)	Segment notes (M/M/1; petrol station M/M/c with costs; machine breakdowns); demos 1–3 contain them	Staff (originals)
6 Further Reading	`Useful links.txt`	J. C. Kantor's SimPy introduction	Optional

(b) Teaching order

(Documented: `5 Source Files/Staff Notes/Session Outline.md` gives the sequence and timings; the notebooks map one-to-one onto it.)

1. 0–30 min: DES and SimPy basics. Show `Simpy Basics.pdf`, then run `1 Intro to SimPy and the M-M-1 queue`.
2. 30–50 min: practical, `Practical - M-M-1 Queue` (vary utilisation; it starts at ρ = 0.83). Break.
3. 60–90 min: run `2 Petrol station (M-M-c) and costs`, including the cost analysis; students try different pump numbers.
4. 90–110 min: run `3 Machine breakdowns (pre-emption)`; `4 … full demo` if time allows.
5. Tutorial (practical session or self-study): `Tutorial Questions.pdf` Q1–Q5. Release the Q1–Q5 notebooks afterwards.

(c) Prerequisites

- S2 Week 02: M/M/1 formulas (Wq = λ/(μ(μ−λ)), W = 1/(μ−λ)), Kendall notation, costs.
- Python generators (`yield`) — introduce briefly; S1 notebooks give general Python practice.
- Open: JupyterLite Week 03 folder; run each notebook's first cell (`%pip install simpy`).

(d) Pitfalls

- The session outline says "Google Colab"; use the JupyterLite notebooks instead.
- Simulated averages vary with the seed and run length. Compare with the Week 02 formulas, not with another student's numbers.
- Pre-emption: a breakdown can interrupt any process holding the machine, including maintenance, and an interrupted job must request the machine again. Demo 4 (not the original `segment_3_demo.py`) handles both; students' own code will crash with an uncaught `Interrupt` unless they catch it.
- Q2 and Q3 answers now end with a replication check: Q2's best is 5 lanes (not 4); Q3 needs 3/4 agents on average, 4/5 to be safe. Q5 utilisation is 75–30%. The original `.py` scripts keep the old answers.
- *(fixed 12 Oct 2026)* Q3 and Q5 notebooks echoed a stray `(0.0, 105.0)` after a plot.

Semester 2 · Week 04 – Problem Structuring and Simulation Build

Handbook topic: *Agent Based Modelling* (S2 week 4) — **mismatch**. This folder holds a problem-structuring workshop with a live SimPy build; Agent Based Modelling is in the Week 06 folder (the Handbook puts the Beer Game in week 6).

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Problem Structuring and Modelling - Emergency Department.pdf`	ED workshop slides (slides 1–2, 9–13, 15–27; slide 14 is a blank whiteboard): scenario, Parts A–D, results	Core
1 Lecture	`Problem Structuring and Modelling - Emergency Department - slides 1-8 (Soft OR introduction).pdf`	Slides 1–8 of the same deck (Soft OR introduction); was called "Session Summary"	Core (shown first)
1 Lecture	`… Emergency Department (annotated).pdf`	All 27 slides with annotations	Optional
1 Lecture	`GORS - Intro to Soft OR and Problem Structuring Exercise.pdf`	Alternative group exercise (government office efficiency, backup tasks)	Optional / alternative session
2 Lecture Demos	`1 Building the ED simulation in stages.ipynb`	The live build: basic queue, triage, patient types, priority, pre-emption, staffing experiments	Core
2 Lecture Demos	`2 Emergency department simulation (full model).ipynb`	Finished model: 2–5 doctors, 40 replications, 95% CIs, costs	Core
3 Tutorial	`Problem Structuring Worksheet.pdf`	Current worksheet (Parts A–E; the possible answers cover A–C)	Core
4 Tutorial Answers	`Problem Structuring Worksheet - Possible Answers.pdf`	Model answers (match the PDF worksheet)	Core, after the session
5 Source Files	`.pptx`, `.docx`, `Python/hospital_sim.py`, `Staff Notes/Session_plan.md`, `Staff Notes/Building the simulation live.md`	Sources; minute-by-minute plan; live-build script	Staff

(b) Teaching order

(Documented: `Session_plan.md` gives a 120-minute plan and `Building the simulation live.md` the six build stages; notebook 1 follows those stages.)

1. 0–10 min: framing — slides 1–2, then the Soft OR intro (the *slides 1-8* PDF; the intro is slides 3–8).
2. 10–30 min: Part A problem structuring (ED slides 10–13) with the worksheet PDF.
3. 30–40 min: Part B predictions (slides 15–16). 40–50 min: conceptual model (slide 17).
4. 50–105 min: Part C live build in `1 Building the ED simulation in stages.ipynb`: stage 1 basic queue, 2 triage, 3 patient types, 4 priority, 5 pre-emption (slides 18–24).
5. 105–120 min: Part D results and discussion (slides 25–27), staffing experiments in notebook 1 (stage 6) and 2 `Emergency department simulation (full model).ipynb`.
6. Release the possible-answers PDF afterwards. The GORS exercise is an alternative or backup session. *(Judgement.)*

(c) Prerequisites

- S2 Week 03: SimPy resources, priority and pre-emptive resources.
- S2 Weeks 01–02: queueing vocabulary.
- Open: JupyterLite Week 04 folder (first cell installs SimPy).

(d) Pitfalls

- *(fixed 12 Oct 2026)* An earlier-draft `.md` worksheet with different questions sat next to the PDF in `3 Tutorial`; it is now in `5 Source Files/Previous Versions`. Use the PDF.
- The session plan's Part A numbering (A1 decisions, A2 questions, A3 factors, A4 sketch) does not match the current worksheet (A1–A6); adjust the plan.
- Both ED PDFs predate the deck's last edit (26 Feb 2026); re-export from the .pptx if it changes again.
- The session plan says not to mention triage or priority before students suggest them, so don't show the full notebook in advance.

Semester 2 · Week 05 – Coursework Part B Release

Handbook topic: *Coursework 2 Release* (S2 week 5) — matches.

(a) Contents

Folder	File	Purpose	Core / optional
6 Further Reading	`How-Vodafone-Uses-Optimization.pdf`	Industry case study (also in S1 Week 06)	Optional
—	`README.md`	Points to `Assignments/02-coursework-part-b` (out of scope of this audit)	—

(b) Teaching order

1. Release and walk through the coursework brief (in `Assignments/`).
2. Suggested reading: the Vodafone case study. *(Judgement.)*

(c) Prerequisites

S2 Weeks 01–04 (simulation, queueing, SimPy) and S1 optimisation, depending on the brief.

(d) Pitfalls

- No lecture material in this folder. The brief lives in `Assignments/`, which the README notes is published with everything else in `content/`; keep staff-only parts out.

Semester 2 · Week 06 – Agent Based Modelling

Handbook topic: week 6 is *The Beer Game*; *Agent Based Modelling* is week 4 — **order mismatch** (see the cross-week check).

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Agent Based Modelling with Mesa - Slides.html`	31-section HTML slide deck: ABM concepts, Game of Life, Schelling, SIR, traffic, wealth/Gini, Mesa	Core
2 Lecture Demos	`1 Game of Life.ipynb` … `5 SIR epidemic.ipynb`	One Mesa notebook per model on the slides (slides 8, 11–14, 15–19, 20–23, 24–27)	Core

(b) Teaching order

1. Work through the HTML slides in order (open with *HTML Viewer* or download).
2. Run `1 Game of Life` → `5 SIR epidemic` in `2 Lecture Demos` as each model comes up. The py.cafe Mesa examples and the Colab notebook linked from the slides are external extras.

(c) Prerequisites

- Python basics and the S2 simulation weeks (stochastic simulation, SimPy processes).
- Internet access for py.cafe / Colab; on the site, each notebook's first cell installs Mesa (about half a minute).

(d) Pitfalls

- *(fixed 19 Oct 2026)* The missing SIR image on slide 26 was replaced by an illustrative chart from the SIR notebook.
- No practical or answers are provided.

Semester 2 · Week 10 – Revision

Handbook topic: week 10 is *Decisions under Uncertainty*; *In-class Test Revision* is week 12 — **mismatch**.

(a) Contents

Folder	File	Purpose	Core / optional
1 Lecture	`Revision.pdf`	7-page revision question sheet	Core
3 Tutorial	`Week 10 Revision Selected Questions.pdf`	Typeset selected questions (GreenGrid, Hock-Brooke, library, F.B. Knight, N&F forecasting)	Core
4 Tutorial Answers	`… - Solutions.pdf`	Worked solutions	Core, after the session
4 Tutorial Answers	`GreenGrid - project selection (Pyomo).ipynb`	Q1: {B, D, E}, benefit 46	Core
4 Tutorial Answers	`University library - simulation (SimPy).ipynb`	Q3: baseline 41 served, 95.12% under 8 min	Core
4 Tutorial Answers	`F.B. Knight - transhipment (Pyomo).ipynb`	Q4: 183 / 247 / 228 / 216 / 181	Core
4 Tutorial Answers	`Hock-Brooke - warehouse simulation.ipynb`	Q2 (corrected): best about 15 spaces, ≈ £221.56/week	Core
4 Tutorial Answers	`Questions 2 and 5 - forecasting and simulation.xlsx`	Q2 simulation sheet (corrected), Q5 forecasting	Core
5 Source Files	`LaTeX/`, `Python/`	Sources (LaTeX for both PDFs, the original scripts)	Staff

(b) Teaching order

(Judgement: follows the question order in the selected-questions sheet.)

1. Hand out `Revision.pdf` / the selected questions; students attempt Q1–Q5 in groups.
2. Go through Q1 (binary project selection, links to S1 Week 05), Q2 (Excel simulation, S2 Week 01), Q3 (SimPy, S2 Week 03), Q4 (transportation/transhipment, S1 Week 08 handbook topic), Q5 (forecasting).
3. Release the solutions PDF and the four notebooks plus the workbook. More revision material is in `Assignments/03-in-class-test/revision`.

(c) Prerequisites

- All of S1 (LP/MILP, transhipment) and S2 Weeks 01–03 (simulation, SimPy). Q5 needs forecasting (moving averages, seasonal indices), which has **no lecture folder** in either repository (Handbook S2 weeks 8–9).
- Open: JupyterLite Week 10 folder; Excel for Q2/Q5.

(d) Pitfalls

- *(fixed 15 Oct 2026)* Q2 was rebuilt with the right data: best about 15 spaces (≈ £221.56/week over 52 weeks), not 8. The workbook's 12-week table still changes with `RAND()`; compare the expected-cost table.
- *(fixed 15 Oct 2026)* The library notebook now prints the Q3(c) table for 2/3/4 advisers with 200 replications; the answer is 4 advisers, not 3.
- Q4 has alternative optimal plans for 183 and 247; any plan with those costs is correct.
- Forecasting (Q5) is examined but not taught in any folder; check the Handbook schedule.

Cross-week order check against the Module Handbook (2025-26)

Source: `content/Handbook/Module Handbook Operational Research (5CM522) 2526.pdf`, *Schedule of work*.

Semester 1

Week	Handbook topic	Folder in `Lectures/Semester 1`	Match
1	Introduction to Operations Research	Week 01 - Introduction to Operational Research	✔
2	Linear Programming 1	Week 02 - Linear Programming 1	✔
3	Linear Programming 2	Week 03 - Linear Programming 2	✔
4	Mathematical Programming – Modelling and Some Issues	Week 04 - Mathematical Programming - Modelling	✔
5	Mixed Integer Programming	Week 05 - Mixed Integer Programming	✔
6	Integer Linear Programming – Further Problems	Week 06 - Integer Linear Programming - Further Problems	✔
7	Scheduling	Week 07 - Flow Shop and Job Shop Scheduling	✔ (holds **two** sessions: flow shop and job shop)
8	Network Models: Transportation and Transhipment Problem	Week 08 - Networks - Shortest Paths and MST	✘ **different topic**
9	Travelling Salesman Problem	Week 09 - Travelling Salesman Problem	✔
10	Vehicle Routing Problem	Week 10 - Vehicle Routing Problem	✔
11	Inventory Modelling	Week 11 - Floor Planning	✘ **different topic**; no inventory lecture material (only older EOQ questions in the test revision papers)
12	Review	—	✘ no folder

Semester 2

Week	Handbook topic	Folder in `Lectures/Semester 2`	Match
1	Introduction to Simulation	Week 01 - Introduction to Simulation	✔ (also holds a min-cost-flow example that belongs to S1 W8)
2	Queuing Theory	Week 02 - Queueing Theory	✔
3	Discrete-Event Simulation	Week 03 - Discrete Event Simulation with SimPy	✔
4	Agent Based Modelling	Week 04 - Problem Structuring and Simulation Build	✘ ABM is in Week 06
5	Coursework 2 Release	Week 05 - Coursework Part B Release	✔
6	The Beer Game	Week 06 - Agent Based Modelling	✘ no Beer Game material
7	From Beer Game to Systems Dynamic Modelling	—	✘ no folder

Week	Handbook topic	Folder in `Lectures/Semester 2`	Match
8	Forecasting	—	✘ no folder
9	Times Series	—	✘ no folder
10	Decisions under Uncertainty	Week 10 - Revision	✘ revision instead
11	Predict Then Optimise: ML + Optimisation	—	✘ no folder
12	In-class Test Revision	(Week 10 - Revision)	revision material sits in week 10

Prerequisite flow: does each week only rely on earlier weeks?

Dependency	Status
S1 W3 Pyomo demos 2–4 rebuild the Week 2 diet problem	OK. Week 2 comes first. The slides call it "Tutorial Question 5a–c", but this year's Week 2 sheet calls it **Q6**. Tell students.
S1 W4–W6 build on W3 Pyomo syntax and W5 binaries	OK, in order.
S1 W7 job shop (Pyomo, disjunctive big-M) needs W5 binaries and big-M	OK.
S1 W9 → W10 (MTZ re-used in VRP)	OK.
S1 W11 floor planning needs big-M non-overlap (W5–W7)	OK.
Transhipment / min-cost flow (Handbook S1 W8)	**Gap.** S1 W8 teaches shortest paths and MST instead. The only transhipment lecture material is in S2 W1 (production-distribution, optional) and S2 W10 Q4 (F.B. Knight, revision); the in-class-test revision papers add practice questions. If S1 W8 is kept as networks, teach the S2 W1 production-distribution deck and notebook in S1 W8 or S1 W12.
S2 W2 queueing formulas → S2 W3 SimPy checks	OK.
S2 W3 segment 3 (priority, pre-emption) → S2 W4 ED live build	OK. W4 needs `PreemptiveResource` and priorities from W3 (demos 3–4, Q4–Q5); `PriorityResource` itself is introduced in W4 stage 4.
S2 W4 (structuring) before S2 W6 (ABM)	Works pedagogically, but the Handbook has ABM in W4. Update the Handbook, or swap the folders.
Forecasting (Handbook S2 W8–9)	**Gap.** S2 W10 revision Q5 (moving averages, seasonal indices, trend) examines forecasting, but no forecasting lecture exists in either repository.
Inventory modelling (S1 W11), Beer Game / system dynamics (S2 W6–7), decisions under uncertainty (S2 W10), predict-then-optimise (S2 W11)	No lecture material in either repository (inventory appears only as older EOQ questions in the test revision papers). These are not losses from the restructure: ORIGINAL had none either.

Recommendation: align the Handbook schedule with the folders, or the folders with the Handbook, before the next delivery. The two content gaps that affect assessment are **transhipment** and **forecasting**: both appear in the S2 W10 revision questions, but neither is taught in its scheduled week.